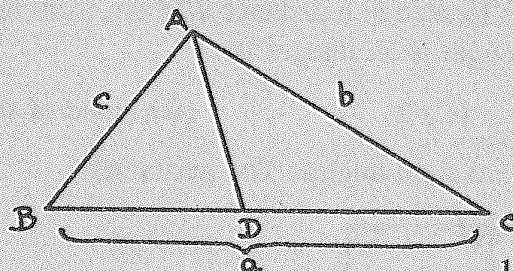


SCREENING TEST—JUNIOR LEVEL 1993

ITMA-93

J

- The ratio of two numbers, if their sum is twice their difference, is
(A) 3 : 1 (B) 2 : 1 (C) 4 : 1 (D) 1 : 2
- In a rose plant a little girl observed that the flowers doubled everyday and in four days she plucked 240 flowers in all. The number of flowers she plucked on the third day is
(A) 128 (B) 120 (C) 112 (D) 64
- The g.c.d. of a and 140 is 35 and their l.c.m. is 420. Then the g.c.d. and l.c.m. of a and 210 are respectively
(A) 35 and 420 (B) 105 and 210 (C) 70 and 210 (D) 105 and 420
- In the adjoining figure the area of the triangle is $\frac{1}{2}bc$ sq. units and AD is a median. Then



- (A) $m\angle BAD = m\angle CAD$ (B) $m\angle ABC = \frac{1}{2}m\angle BDA$
 (C) $m\angle ABD = m\angle ACD$ (D) $m\angle ABC = \frac{1}{2}m\angle ADC$
- The median of the data given below is 50.

Marks	23	35	40	50	55	60	20
Frequency	7	8	12	10	x	10	11

Then,

- (A) $16 \leq x \leq 37$ (B) $x > 37$ (C) $0 \leq x \leq 16$ (D) $x \geq 16$
- $a^4 - b^4 = p_1 \times p_2$, $a, b, p_1, p_2 \in \text{IN}$ (the set of all natural numbers), $a > b, p_1 > p_2, p_1, p_2$ are prime numbers. Then
 (A) $a^2 + b^2 = p_1$ and $(a - b) = p_2$ (B) $a^2 + b^2 = p_2$ and $(a - b) = p_1$
 (C) $a^2 + b^2 = p_1$ and $(a + b) = p_2$ (D) $a^2 + b^2 = p_2$ and $(a + b) = p_1$
 - ABC is a triangle with measure of angles, A , and B 30° and 60° respectively and $AB = 10$ cm. The length of the shorter trisector of $\angle C$ is
 (A) $5\sqrt{3}$ cm (B) $\frac{5}{\sqrt{3}}$ cm (C) $\frac{5}{2\sqrt{3}}$ cm (D) $\frac{5}{2}\sqrt{3}$ cm

8. A man born in the first half of the nineteenth century was x years old in the year x^2 . He was born in

- (A) 1849 (B) 1825 (C) 1813 (D) 1806

9. Two candles of the same height are lighted simultaneously. The first burns completely in 2 hours and the second in 1 hour 15 minutes. Each burns at a constant rate. The time in hours after which the first candle is thrice the height of the second candle is

- (A) $\frac{19}{20}$ (B) $\frac{20}{19}$ (C) $\frac{7}{8}$ (D) $\frac{8}{7}$

10. The number of values of n for which $2^{11} + 2^8 + 2^n$ is a perfect square is

- (A) none (B) 1 (C) 2 (D) infinite

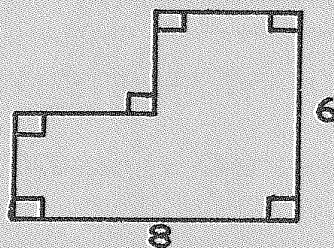
11. The number of ordered pairs (a, b) for which $a2b308$ is divisible by 33 is

- (A) none (B) 1 (C) 2 (D) many

12. In a circle of radius 13 cm a point P is located at a distance of 5 cm from the centre. The number of chords passing through P whose length is an integer is

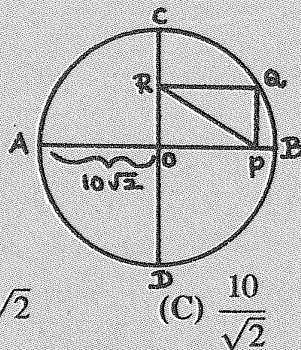
- (A) 4 (B) 6 (C) 2 (D) None of these

13. In the adjoining figure all the angles are right angles. The perimeter



- (A) can not be found as the data is not sufficient (B) is found to be 14
(C) is found to be 28 (D) None of the above

14. In the adjoining figure AB and CD are perpendicular diameters, and $OPQR$ is a rectangle. Then PR measures



- (A) 10 (B) $10\sqrt{2}$ (C) $\frac{10}{\sqrt{2}}$ (D) None of these

15. The minimum value that the polynomial $1 + x - x^2$ reaches is

- (A) 1 (B) 0 (C) $\frac{5}{4}$ (D) not existing

16. In a square of side 10 cms, another square of side 20 cms, hinged at the centre of the first square is rotated. The area common to both the squares

- (A) does not change
- (B) increases when rotating in the clockwise direction
- (C) increases while rotating in the anticlockwise direction
- (D) increases and then decreases

17. In a cylinder with height 10 cms and radius of the base 7 cms, a point P on the circumference of the base and a point Q vertically above are connected along the surface by a tight inelastic string which goes round the cylinder exactly once. Then the length of the string is

- (A) 14 cm
- (B) 10 cm
- (C) $\sqrt{196\pi^2 + 100}$
- (D) between 10 cm and 20 cm

18. Five letters are put in 5 addressed envelopes. The number of ways in which only four of the letters are put in the correct envelopes is

- (A) 3
- (B) 2
- (C) 1
- (D) None of these

19. If
$$\begin{array}{r} U \quad S \quad A \quad + \\ U \quad S \quad S \quad R \\ \hline P \quad E \quad A \quad C \quad E \end{array}$$
 and if each letter stands for a different digit, then A is

- (A) 2
- (B) 3
- (C) 8
- (D) 5

20. The smallest positive integer which cannot be the exterior angle in degrees of a regular polygon is

- (A) 1
- (B) 2
- (C) 4
- (D) 7

21. An ordered triplet (a, b, c) , where a, b, c are nonzero real numbers is said to be good if each is a product of the other two. The number of such good triplets is

- (A) 1
- (B) 4
- (C) 5
- (D) 8

22. The tens digit of the number $1! + 2! + 3! + \dots + 1993!$, if $n!$ is defined to be $1 \times 2 \times 3 \times \dots \times n$, is

- (A) 1
- (B) 2
- (C) 3
- (D) 5

23. Four planes pass through a common point, but no three contains the same line. The number of regions into which the whole space is divided by these planes are

- (A) 10
- (B) 8
- (C) 12
- (D) 14

24. The minimum number of straight lines we need to make exactly 100 squares is

- (A) 12
- (B) 14
- (C) 15
- (D) None of these

25. The number of values of n for which the sum of the squares of the first n natural numbers is a perfect square is

- (A) 2 (B) 5 (C) 0 (D) None of these

26. The number of positive integers n such that $n = 6[\sqrt[3]{n}] + 1$, where $[x]$ is the greatest integer $\leq x$, is

- (A) 4 (B) 3 (C) 1 (D) 2

27. In the set of all positive rationals Q^+ , a binary operation $*$, having the following properties, is defined: (i) $(x * y) \times (z * t) = (xz) * (yt)$, (ii) $x * x = 1$ (iii) $x * 1 = x$. Then $54 * 27$ is

- (A) 3 (B) 2 (C) 1 (D) 0

28. A rectangular floor is composed of square tiles of the same size, 40 on one side and 36 on the other. If a straight line is drawn diagonally across the floor, corner to corner, the number of tiles it crosses is

- (A) 36 (B) 40 (C) 72 (D) 76

29. $\sqrt{3 + 2\sqrt{2}} - \sqrt{3 - 2\sqrt{2}}$ is equal to

- (A) $\sqrt{2}$ (B) $2\sqrt{3}$ (C) $4\sqrt{2}$ (D) 2

30. In triangle ABC , $\angle B$ is obtuse, $AB = 6$, $BC = 8$; then

- (A) $8 < AC < 10$ (B) $8 < AC < 12$ (C) $10 < AC < 12$ (D) $10 < AC < 14$